

UNIT 1

Managing yourself

Every man is the architect of his own fortune.

—Appius Claudius Caecus



In this unit, you are going to:

- read an excerpt from a bestselling book about habits;
- write an article about self-confidence;
- read a lecture transcript about time management;
- present a talk show about learning efficiency.

Welcome to the unit

It is often believed that habits can determine our level of success. Read the list of good habits below and discuss the following questions in pairs.

- ☐ I am always punctual for appointments.
- ☐ I never put off what I can do today until tomorrow.
- ☐ I always finish what I have started, even when it gets difficult.
- ☐ I always do the things I *need* to do before doing the things I *want* to do.
- ☐ I always reflect on the mistakes I have made in order to learn from them.
- ☐ I always practise a lot to excel in what I set out to do.

- 1 Which good habits in the list do you have? Tick the boxes above.
- 2 What other good habits do you have or want to develop?
- 3 What bad habits do you need to give up? How do you plan to do it?

Reading



Good habits are essential for an individual to grow and develop. The excerpt below is taken from Stephen Covey's bestselling book *The 7 Habits of Highly Effective People*. Before you read the excerpt, think about the following questions:

- Why do you think habits are important to us?
- What habits can help us become highly effective people?

The 7 Habits of Highly Effective People (Excerpt)



The 7 Habits—An Overview

We are what we repeatedly do. Excellence, then, is not an act, but a habit.

—Aristotle

- Our character, basically, is a composite of our habits. “Sow a thought, reap an action;
5 sow an action, reap a habit; sow a habit, reap a character; sow a character, reap a destiny,” the maxim goes.

Habits are powerful factors in our lives. Because they are consistent, often unconscious patterns, they constantly, daily, express our character and produce our effectiveness ... or ineffectiveness.

- 10 As Horace Mann, the great educator, once said, “Habits are like a cable. We weave a strand of it every day and soon it cannot be broken.” I personally do not agree with the last part of his expression. I know they can be broken. Habits can be learned and unlearned. But I also know it isn’t a quick fix. It involves a process and a tremendous commitment.
- 15 Those of us who watched the lunar voyage of *Apollo 11* were transfixed as we saw the first men walk on the moon and return to earth. Superlatives such as “fantastic” and “incredible” were inadequate to describe those eventful days. But to get there, those astronauts literally had to break out of the tremendous gravity pull of the earth. More energy was spent in the first few minutes of lift-off, in the first few miles of travel, than
20 was used over the next several days to travel half a million miles.

- Habits, too, have tremendous gravity pull—more than most people realize or would admit. Breaking deeply imbedded habitual tendencies such as procrastination, impatience, criticalness, or selfishness that violate basic principles of human effectiveness involves more than a little willpower and a few minor changes in our
25 lives. “Lift-off” takes a tremendous effort, but once we break out of the gravity pull, our freedom takes on a whole new dimension.

Like any natural force, the gravity pull can work with us or against us. The gravity pull of some of our habits may currently be keeping us from going where we want to go. But it is also the gravity pull that keeps our world together, that keeps the planets in their orbits and our universe in order. It is a powerful force, and if we use it effectively, we can use the gravity pull of habit to create the cohesiveness and order necessary to establish effectiveness in our lives.

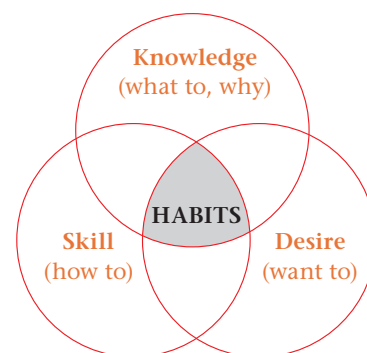
“Habits” Defined

For our purposes, we will define a habit as the intersection of *knowledge*, *skill*, and *desire*. Knowledge is the theoretical paradigm, the *what to do* and the *why*. Skill is the *how to do*. And desire is the motivation, the *want to do*. In order to make something a habit in our lives, we have to have all three.

I may be ineffective in my interactions with my work associates, my spouse, or my children because I constantly tell them what I think, but I never really listen to them. Unless I search out correct principles of human interaction, I may not even *know* I need to listen.

Even if I do know that in order to interact effectively with others I really need to listen to them, I may not have the skill. I may not know *how* to really listen deeply to another human being.

But knowing I need to listen and knowing how to listen is not enough. Unless I *want* to listen, unless I have the desire, it won’t be a habit in my life. Creating a habit requires work in all three dimensions.



The being/seeing change is an upward process—being changes seeing, which in turn changes being, and so forth, as we move in an upward spiral of growth. By working on knowledge, skill, and desire, we can break through to new levels of personal and interpersonal effectiveness as we break with old paradigms that may have been a source of pseudo-security for years.

EFFECTIVE HABITS
Internalized principles and
patterns of behavior

It’s sometimes a painful process. It’s a change that has to be motivated by a higher purpose, by the willingness to subordinate what you think you want now to what you want later. But this process produces happiness, which can be defined, in part at least, as the fruit of the desire and ability to sacrifice what we want *now* for what we want *eventually*.



A Understanding the text

A1 Read the excerpt and complete the chart below.

Paras. 1–2	⇒	Habits are powerful factors in our lives.
Para. 3	⇒	Both learning and unlearning a habit involve (1) _____.
Paras. 4–6	⇒	The gravity pull of habit can be used to (2) _____.
Paras. 7–10	⇒	A habit is the intersection of (3) _____.
Paras. 11–12	⇒	Habits allow us to move in (4) _____.

A2 Read the excerpt again carefully and answer the following questions.

1 Where does the author disagree with Horace Mann?

2 What is the similarity between launching spacecraft and breaking old habits?

3 What does creating a habit require?

4 Why is the being/seeing change sometimes a painful process?

A3 In pairs, discuss the following questions.

1 How does the author illustrate the importance of habits? Use details from the excerpt to support your opinion.

2 The author encourages people to always seek to change and improve by moving in an “upward spiral of growth”. How do you understand this “upward spiral of growth”?

3 What good habits do you have? Share one of them with your classmates and describe how you successfully developed it. Use the author’s definition of habits to help you.

B Building your language

B1 Emily has found a passage explaining the seven habits put forward by Stephen Covey. Complete the passage with the correct forms of the words and phrases in the box below.

interaction
tendency

commitment
excellence

sow
inadequate

in turn
search out

The 7 Habits of Highly Effective People is a bestseller written by Stephen Covey. In this book, he proposes that to ⁽¹⁾ _____ the seeds of success, we should move through three stages of maturity: dependence, independence and interdependence. In order to grow and develop as we progress through this process, Covey identifies seven habits.

To move from being dependent upon others to becoming independent requires the development of three essential habits. The first of these is to be proactive. That is, we should resist the ⁽²⁾ _____ to wait for things to happen; instead, we should make things happen. Similarly, we must not wait for a problem to develop, but rather try to predict what the problem might be and then do something to prevent it from happening. The second habit is to visualize what we want to achieve so that we can think about how to go about achieving it. The third habit is to list your tasks in order of priority. Covey advises that we should make a(n) ⁽³⁾ _____ to doing what we have to do before doing what we would like to do.

To become interdependent requires three habits that relate to the ⁽⁴⁾ _____ with others. Firstly, we need ⁽⁵⁾ _____ solutions which allow us as well as our associates to experience success. Secondly, we should develop the ability to listen to and empathize with others. Essentially, we must first seek to understand others so that, ⁽⁶⁾ _____, we can be understood. Thirdly, we must understand the importance of teamwork, which helps us achieve those goals that we could not achieve on our own.

Just reaching a state of interdependence is ⁽⁷⁾ _____—we must still continue to improve ourselves. The last habit is to sharpen the saw, which surrounds all the other habits because it is the habit that makes the others possible. It is about renewing and enhancing the greatest property we have—us. Investing in our own education is essential if we are to progress in an upward spiral and achieve continued ⁽⁸⁾ _____.

B2 The author uses quotes to explain his idea of habits. Find the quotes in the excerpt, write a short paragraph about habits and support your idea with a quote. You can use the following quotes to help you.

The unfortunate thing about this world is that good habits are so much easier to give up than bad ones.

—William Somerset Maugham

In essence, if we want to direct our lives, we must take control of our consistent actions. It's not what we do once in a while that shapes our lives, but what we do consistently.

—Tony Robbins

Grammar and composition

Simile and metaphor

A Exploring the rules

Below is a magazine article about the fast pace of modern life. Find the similes and metaphors, and fill in the table below.

All the world is a racetrack, and all the men and women are cars moving faster and faster. Old sayings such as “Anything worth having is worth waiting for” used to carry important messages; however, their impact has been reduced largely due to continued improvements in technology. Our computers and mobile phones can now be fired up in milliseconds and predictive text can guess what we want to say even before we have had time to organize our thoughts. Emails and text messages travel the world in an instant, and not surprisingly, replies are expected immediately, 24/7. New models of electronic devices, with faster processing speeds, come onto the market at increasingly short intervals. Indeed, our modern life is just like motor racing, where we constantly expect to overtake the car in front of us.

Unfortunately, this fast pace of life is having a negative impact on our physical and mental well-being. Studies have shown that a fast-paced lifestyle typically keeps us as busy as a bee, leaving less time for exercise. Over the long term, we may suffer from sleep problems, obesity and even heart attacks. Our relentless struggle to get things done as soon as possible also brings with it mental problems such as anxiety and depression. Unless we recognize the problem, it is only a matter of time before the wheels fall off.

Similes	Indeed, our modern life is just like motor racing, ...
Metaphors	All the world is a racetrack, and all the men and women are cars moving faster and faster.

Working out the rules

- Both simile and metaphor are figures of speech that draw a (1) _____ (contrast/comparison) between two things. Using simile or metaphor often helps explain something abstract or unfamiliar more clearly, or describe it more vividly.
- Simile uses the words *like* or (2) _____, while metaphor does NOT use these words.

👉 Grammar and composition notes → page 91

B Applying the rules

B1 Decide if each of the following sentences is a simile or a metaphor. Explain what things are being compared and how they are alike.

1 The sun was a toddler insistently refusing to go to bed: it was past eight thirty and still light.

2 Life is like riding a bicycle.

3 Eyes are the windows to the soul.

4 Without glasses I am as blind as a bat.

5 He loved to camp, so his job was like a holiday for him.

B2 Below is a magazine article about how to slow down our pace of life. Complete it with the phrases in the box below. Write the letters in the blanks.

a a pressure cooker

c recharge our batteries

b hamsters in an exercise wheel

d get off the exercise wheel

Nowadays we are living in the fast lane. Just like ⁽¹⁾ _____, we should constantly run simply to stay in the same place. Some people think that if we push ourselves to the limit now, we can then live a wonderful life and ultimately slow down, relax and enjoy life itself. Unfortunately, if we always live in ⁽²⁾ _____, we will eventually burn ourselves out, ending up in a living nightmare.

For those of us weighed down with pressure, how can we ⁽³⁾ _____?

Firstly, we must learn to say no, both to ourselves and to others, if we feel we are doing too much. We need to recognize our limitations and focus on the things that really matter to us. This way, we can add more white space to our schedule. Secondly, it is important that we should focus on one thing at a time. Jumping back and forth between multiple tasks makes us feel rushed and stressed, and ultimately we cannot complete each task particularly well. Finally, we should always find the time to do something that we like, such as reading a favourite book or walking in nature. Doing things for fun and relaxation, especially after a long period of work or study, not only directs our attention away from the things that have been a source of pressure, but also allows us to ⁽⁴⁾ _____.

B3 Have you been in a situation where you were under pressure from a fast-paced lifestyle? How did you improve the situation? Write a short paragraph about this experience. Use at least a simile and a metaphor.

Integrated skills

Building up your self-confidence

A David is talking to his coach after losing a school basketball game. Listen and finish the exercises below.



A1 Listen to the conversation and choose the best answer. Circle the letters of the correct options.

- 1 Why is David upset?
 - a He doesn't like basketball.
 - b He missed the game-winning shot.
 - c He has no time to himself.
- 2 Why does the coach mention Michael Jordan?
 - a To encourage David.
 - b To teach David how to shoot.
 - c To praise Jordan for his contributions.
- 3 Where does this conversation probably take place?
 - a In the school gym.
 - b In the teacher's office.
 - c In the school dining hall.



A2 Listen to the conversation again and complete the notes below.



The coach's advice

How to stop feeling upset

- It's normal to fail. Michael Jordan lost almost ⁽¹⁾ _____ games and missed more than 9,000 shots. ⁽²⁾ _____, he missed the game-winning shot.
- It's much more important to keep trying to ⁽³⁾ _____.

How to ⁽⁴⁾ _____

- Don't beat yourself up when you get something wrong, for confidence grows when you ⁽⁵⁾ _____.
- Self-doubt is OK, but your performance will improve if you don't let the worries ⁽⁶⁾ _____.
- Do more practice so that you can do better in the next match.

B David has found an article about causes of low self-confidence and ways to build up one's self-confidence. Read the article and fill in the table below.

Build up your self-confidence

Self-confidence is about trusting your judgement and your ability to do something well. It is also about being willing to consider the possibility that you are wrong and to change your mind. It is a combination of courage and humility, which leads to open-minded thinking. Low self-confidence, in contrast, will make you feel unsuccessful, unmotivated and inferior to others. It often prevents you from doing your best and perhaps from even trying. When you do try, your lack of trust in yourself may ultimately lead you to fail.

The reasons for low self-confidence are many and varied, and often differ from person to person. One of the reasons is the home environment where children grow and develop. They may find it difficult to develop self-confidence if their parents are critical or overprotective. Once they fail to fulfil expectations, they may feel bad about themselves and get trapped in the cage of self-doubt. Negative experiences also have a huge impact on self-confidence. If a child gets bullied in school or in the neighbourhood, he or she may be afraid of speaking in public, even when he or she grows up to be an adult.

The good news is that your self-confidence can be improved. Firstly, work on positive thinking. A positive attitude will lead to positive results. Do not worry about letting others down. All they expect is that you do your best. Turn your negative thoughts into positive ones. You can always talk to yourself in a positive way, such as “I can do it!” or “I’ll give it another try!”. Secondly, set realistic goals for yourself and visualize yourself achieving these goals. Confidence is built on a sense of accomplishment. It is important to break down the overall goal into smaller and more measurable tasks. For those tasks you are not so sure about, ask your friends or family for advice. Also, understand that it is perfectly normal if you do not succeed in completing a task. Finally, permit yourself to make mistakes. It is not always possible to get things right the first time. Therefore, do not always feel bad about yourself when you make a mistake, but accept it as an opportunity to reflect on what has gone wrong and how you can do better when you try again. Recognize your mistakes and make a plan to avoid them in the future.

Negative effects of low self-confidence	
Causes of low self-confidence	
Ways to improve self-confidence	

C In pairs, have a discussion about self-confidence. Use the following questions to help you.

- Have you experienced something that made you feel less self-confident? If yes, what was it?
- What causes low self-confidence?
- What are the negative effects of low self-confidence?
- How can you build up your self-confidence?

D Write an article about self-confidence. Use your ideas from part C and the information in parts A and B to help you. Check and improve your writing after you finish it.

Title: _____

Introduction to
self-confidence

Causes and
effects of low
self-confidence

Ways to
improve
self-confidence

Conclusion

Extended reading



Read the lecture transcript about time management.



Good afternoon. Let me begin my presentation by asking you this question: how much time does it take you to sleep, eat, commute and maintain your personal hygiene every day? Do some quick calculations, and it will occur to you that these essential activities take up an enormous part of your day, leaving only limited time to stay with
5 family and friends, pursue a hobby or work on personal development.

Managing your time effectively, therefore, becomes a matter of critical importance. It means that you will be able to accomplish much more with less effort in shorter periods of time. It will keep your stress levels to a minimum, boost your performance and give you a sense of accomplishment, leading to improved self-confidence and a more positive
10 outlook on life. Importantly, effective time management will not only support your studies today, but also help you throughout your adult life. So, now that you understand why good time management is of great importance, how can you improve your ability to manage your time? There are many things you can do, but let me share with you what I consider to be the three most important skills.

15 The first thing you need to do is to plan ahead. Get into the habit of thinking about upcoming events and making a monthly, weekly or even daily to-do list. This will give you a measure of control over what is going to happen. Whether it is a planner that you stick on the wall or a calendar app on your smartphone, finding an organizing tool that works well for you is essential. Then, record all important events such as exams, projects
20 and rehearsals, so that you know all the dates and deadlines and can prepare accordingly.

After you have all that needs to be done in mind, you can set about creating favourable conditions to actually get things done. This is about the skill of organization. Being well organized includes making sure you have everything ready for tomorrow's lessons and after-school activities. It is also important for you to organize your personal space

25 at home so that when you are doing your homework, you are in a tidy place, free from distractions. This includes putting away your mobile phone until you have finished everything that needs to be done.

Most importantly, you must learn to prioritize all your tasks. There is only a limited amount of time in a day, and even with good planning and organization, it is not always
30 possible to complete everything that needs to be done. In such circumstances, you will have to decide which task should be completed first. According to Stephen Covey's Time Management Matrix, there are four equal areas indicating the tasks that you have to or want to do every day. You should classify the activities in the four areas and identify which are urgent and important, which are not urgent but important, which are
35 urgent but not important, and which are neither urgent nor important. Understanding that tasks can be classified in this way will help you prioritize what needs to be done. While choosing which task needs to be done first can be a difficult decision to make, it is important not to let peer pressure or the quest for short-term rewards influence you, but to focus on the goals you have set for yourself and stick to them.

40 To bring my presentation to a conclusion, let me point out that once you start taking little steps every day to plan, organize and prioritize the activities you intend to devote your energy, skills and talents to, you will immediately begin to reap the rewards of your actions. The well-known Chinese writer Lu Xun said, "Time is like the water in a sponge. If you are willing to squeeze, you will always get some." Time is a finite
45 resource, but with a little effort you can always find the time to do what needs to be done. To say "I have no time" is only an excuse for avoiding doing something that could actually be done. I hope you have found my talk interesting and useful and that you never again have to say "I don't have the time to do that".

A Write a summary of the lecture transcript. Use the following ideas to help you.

- Importance of time management
- Ways to manage your time effectively

B If you were to give a lecture about time management, what information would you like to add to the lecture transcript?

C Give an example of using your time effectively and share with your classmates.

Project

Presenting a talk show about learning efficiency

A As a class, discuss what tips can be presented to help students learn more efficiently. You can use the ideas below or think of other ideas. Then in groups, choose one tip to research.

using a mind map
forming a study group
taking notes

reflecting on what you have learnt
making a revision plan
understanding your learning style

B As a group, research your chosen tip on learning efficiency. Use the questions below to help you.

- What is the tip?
- Why is it important?
- How can we use it?

C As a group, write the script of your talk show. Use the example below to help you. Then present your talk show to the rest of the class.

Host: Good morning. Today we will be discussing how students can learn more efficiently. My guests on the show are students from a local high school who would like to share some learning tips with you. Welcome to the show.

Student 1: Thank you. It is a pleasure to be here and we have a learning tip which can help all our fellow students.

Host: That sounds great. Do tell us.

Student 1: We have been using mind maps in our studies. A mind map is a diagram that consists of a central idea with related ideas branching out, all arranged in an interconnected manner. It shows not only the individual parts but also the overall structure of the subject.

Host: Can you tell us more about the importance of this tip?

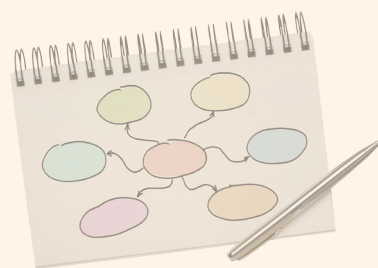
Student 2: It enables you to see the connection between different ideas and encourages you to come up with more ideas. Consequently, it helps you think more logically and creatively.

Host: Does a mind map work for everyone?

Student 3: Yes, but it works particularly well for visual learners since it is designed to be seen. And it is helpful as long as there is information to organize. For example, you can use mind maps to brainstorm ideas, take notes, plan your writing and learn new words.

Host: Can you give us an example?

Student 1: Sure. ...



Assessment

Complete the following learning journal to assess your performance. Then work in groups and exchange your learning journal with your partners.

Learning journal

In this unit I learnt _____

I was most interested in _____

I did well in _____

However, there is something I need to improve on: _____

In order to make more progress, I will _____

* Assess your learning of vocabulary and grammar by doing language practice on pages 57–58.

Further study



Self-esteem is the subjective assessment we make of our own worth or value. Low self-esteem can have a negative impact on our mental health and our overall quality of life, so it is important to feel good about who we are. Watch a film in which someone with low self-esteem finds ways to feel better about themselves.



It is often said that humans are creatures of habit. If we reflect on our daily routine, we will recognize our own habits, both good and bad. Good habits aid our personal growth, whereas bad habits get in the way of our development. Read a book on how to get rid of bad habits and develop good ones.



UNIT 2

Thinking out of the box

Daring ideas are like chessmen moved forward; they may be beaten, but they may start a winning game.

—Johann Wolfgang von Goethe



In this unit, you are going to:

- read a website article about thought experiments;
- write an article about creativity;
- read a magazine article about the invention of eyeglasses;
- make a leaflet about an innovation.

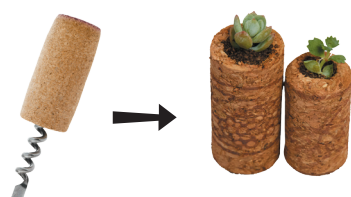
Welcome to the unit

Everyone can come up with creative ideas. Try the tests below to exercise your creative genius and discuss the following questions in pairs.

Alternative use test

Everyday items can have a wide variety of unexpected uses. List as many possible ideas for how to use a cork.

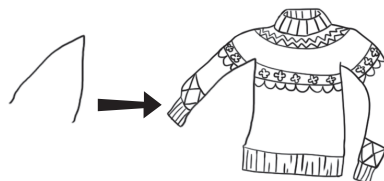
[Example]



Incomplete figure test

Things as simple as a line can open up interesting possibilities for creative minds. Look at the line below, see what picture you can create based on this line and describe your picture.

[Example]



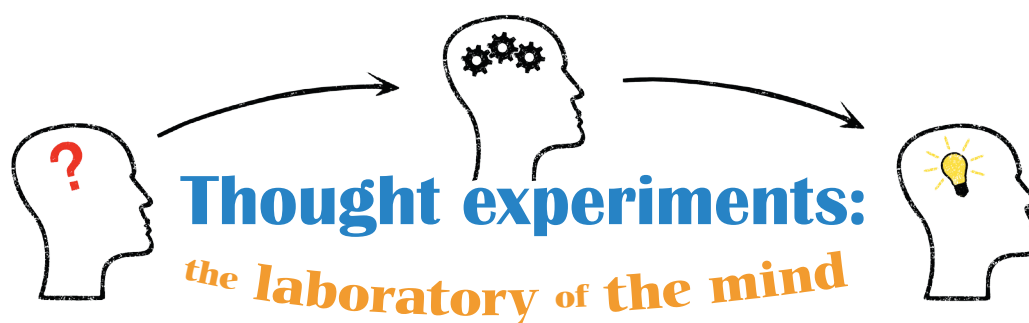
- 1 What other creative answers can you give in the above tests?
- 2 What qualities do you think creative people share?

Reading



Curious by nature, humans explore the world with various experiments. The website article below is about a unique type of experiment known as thought experiments. Before you read the article, think about the following questions:

- What do we need to do in order to conduct an experiment?
- Look at the title and the pictures. What kind of experiment do you think thought experiments are?



What would we observe if we travelled at the speed of light? This is a question Albert Einstein asked himself at the age of 16. Imagining himself pursuing a beam of light, he reasoned that he should observe such a beam of light as an electromagnetic field at rest. This intuitive thinking later played an important role in the development of his special theory of relativity, as Einstein acknowledged in his 1949 *Autobiographical Notes*. It came to be known as one of Einstein's most famous thought experiments.

So, what are thought experiments? Put simply, they are experiments carried out in the laboratory of the mind; in other words, they take place in the imagination. We set up some hypothetical situation, carry out an operation, see what happens, and draw a conclusion from our analysis. As we explore the world in greater depth, we may reach a point where it is impossible to run an experimental scenario in the real world due to physical, ethical or financial limitations. In such circumstances, we move into the area of thought experiments, which function as a valuable means to test the limits of our knowledge and advance our understanding of the world.

While the term "thought experiment" originated in the 19th century, the use of thought experiments can be traced much further back, when ancient philosophers used them to explore difficult questions. About 2,500 years ago, Chinese philosopher Zhuangzi recorded one of the earliest thought experiments: if you cut a long stick in half and then cut the remaining part in two each following day, you will never finish cutting it. With the birth of modern science, thought experiments became an important tool for scientists, especially in physics. Today thought experiments are regularly used in the areas of philosophy, economics and the sciences.

Throughout history, thought experiments have contributed to the development of scientific theories. With logical reasoning, thought experiments are capable of putting

25 forward a new theory and supporting or disproving an existing theory. A classic example
of this is Galileo Galilei's thought experiment with two falling balls. In Galileo's
day, Aristotle's belief that a heavier ball should fall faster than a lighter one was well
accepted. Doubtful, Galileo asked what would happen if the two balls were tied together.
He then figured out two contradictory outcomes: because the two balls fell at different
30 speeds, the lighter ball would slow down the fall of the heavier one; but if the two balls
were considered as a whole, together they weighed more and should fall faster than each
individual ball. Using the power of the mind, Galileo was able to expose the paradox
and prove Aristotle wrong.

However, thought experiments are not always about proving something right or wrong
35 with a definite answer. Instead, by making abstract, complex ideas more concrete and
accessible, thought experiments can stimulate debate and lead to a more thorough
understanding of the topic in question. A famous example is the Ship of Theseus, which
was first written down by the ancient Greek philosopher and biographer Plutarch in the
1st century. He mentioned that for many centuries, the people of Athens kept the old ship
40 of the ancient Greek hero Theseus. Over the years, all the original wooden parts of the
ship were damaged and replaced by the local people. Plutarch questioned whether this
ship was still Theseus's old ship, or whether it was an entirely new one. Many centuries
later, the English philosopher Thomas Hobbes took the thought experiment one step
further: what if someone collected all the original parts and rebuilt another ship? Which
45 of the ships would then be the ship of Theseus? Even today, the Ship of Theseus still
sparks intense debate and continues to raise many questions about the nature of identity.

With hypothetical situations, thought experiments allow us to explore the outcome
of a given hypothesis which has real-life significance. They also help us stretch our
minds by confronting and considering difficult questions. Why not ask our own "What
50 if ..." questions? By adopting new and innovative ways of thinking, we may arrive at
surprising conclusions.



A Understanding the text

A1 Read the website article and complete the chart below with the main idea of each paragraph.

Introduction	Para. 1: Einstein's thought experiment played an important role in the development of his special theory of relativity.
	Para. 2: _____
Main body	Para. 3: _____
	Para. 4: Thought experiments have contributed to the development of scientific theories.
	Para. 5: _____
Conclusion	Para. 6: _____

A2 Read the website article again carefully and answer the following questions.

1 Why is it sometimes impossible to carry out experiments in the real world?

2 In what areas are thought experiments regularly used today?

3 What was the paradox exposed by Galileo?

4 What questions did Thomas Hobbes put forward about the ship of Theseus?

A3 In pairs, discuss the following questions.

1 How do you think thought experiments are different from and similar to real-life experiments?

2 The author gives examples of some famous thought experiments. Which of them is the most impressive to you? Why?

3 What might be the disadvantages of thought experiments? Support your opinion with reasons.

B Building your language

B1 The passage below is about a thought experiment created by Isaac Newton. Complete the passage with the correct forms of the words and phrases in the box below.

means
outcome

depth
accessible

thorough
carry out

be capable of
figure out

We have all heard about gravity. We know that gravity makes apples fall to the ground and stops us from flying off into space. But have you ever wondered, when looking at the Moon, why it does not come crashing down to the Earth? Well, this is a question that also puzzled Isaac Newton. To find the answer, he ⁽¹⁾ _____ a thought experiment which is now known as Newton's Cannon.

In his book published in 1687, Newton imagined shooting a cannonball horizontally from the top of a very high mountain, where there was no air and consequently no air resistance. Without the force of gravity, the cannonball would go in a straight line. However, as the force of gravity pulls everything earthwards, the cannonball would travel in a curve and finally reach the ground. When the cannonball was fired at higher speed, it would travel farther and farther before it hit the ground. Then Newton ⁽²⁾ _____ that at the right speed, the cannonball would follow exactly the curve of the Earth's surface, never falling to the ground. The ⁽³⁾ _____ was that the cannonball successfully went into orbit around the Earth.

The speed required for the cannonball to orbit the Earth is 7.9 kilometres per second. In Newton's day, there was simply no ⁽⁴⁾ _____ to fire a cannonball at that speed. However, Newton constructed this brilliant thought experiment with a clever combination of his imagination and knowledge of physics, which contributed to a more ⁽⁵⁾ _____ understanding of gravity as a universal force. Newton's Cannon also became the first and most ⁽⁶⁾ _____ explanation for orbital motion. It was not until centuries later, when space rockets were invented, that we ⁽⁷⁾ _____ testing Newton's idea in real life. Thanks to Newton's ⁽⁸⁾ _____ of thinking, many satellites, routinely used for communication and navigation, orbit the Earth. Just like the Moon, you now know why they don't fall on your head!

B2 The website article uses an example to introduce the topic of thought experiments, which makes the introduction more appealing and accessible to the reader. Find the example in the article and write a similar introduction to an article about inventions. Use the information below about the innovation of paper to help you.

Inventor	Cai Lun (the Eastern Han Dynasty)
Process	• Letting sheets of wet fibre float in the water • Slowly draining the moisture until the paper was bone dry
Significance	Leading to a rapid increase in paper production

Grammar and composition

Idioms

A Exploring the rules

Below is a blog post about the popularity of makerspaces in today's world. Find the idioms, guess their meanings and fill in the table below.

In the past, people worked with tools and made things from different materials using their hands. However, as modern technology has been advancing in leaps and bounds, it has now become far too easy for us to buy ready-made things, and some people are lacking in this kind of creativity. Fortunately, due to a snowball effect, makerspaces are multiplying, providing places where people can claim back the right to make things with their own hands.

A makerspace is a creative workshop that aims to get back to basics, encouraging young minds to gather together to share ideas, innovate and invent things. Making things together with other people not only adds an element of fun, but also allows makers to bounce ideas off one another. Makerspace equipment can be simple such as workbenches, shelves stocked with art supplies, and bins containing wood and plastics. Some makerspaces may provide sophisticated equipment such as computers, three-D printers and laser cutters. Whatever the level of equipment, all makerspaces are devoted to bringing young people together in the spirit of collaboration, motivating them to experiment with different crafts and technologies.

Of course, failure is part and parcel of the maker experience; however, at makerspaces “fail” just stands for “first attempt in learning”. So why not be an eager beaver and get a few friends together to start making things? Once you do that, you will discover that the creative possibilities are endless!

Idioms	Meanings
in leaps and bounds	very quickly
a snowball effect	
	return to the simplest or most important ideas
part and parcel	

Working out the rules

- An idiom is a fixed expression whose meaning is usually different from the meanings of the individual words.
- Idioms are usually used in _____ (formal/informal) language, and they help make the language vivid and colourful.

👉 Grammar and composition notes → pages 91–92

B Applying the rules

B1 Rewrite the following sentences using the idioms in the brackets.

- 1 In order to create something truly original, we sometimes need to give up complex concepts and return to the simplest or most important ideas. (get back to basics)

- 2 When it comes to writing, it is important to realize that at some point you will experience writer's block. (along the line)

- 3 This simple yet innovative design gave them an advantage over the other competitors. (gain the upper hand)

B2 Below is a personal account of a girl's first experience of making things herself. Complete it with the correct sentences in the box below. There is one sentence you do NOT need to use. Write the letters in the blanks.

I've just got back home from school with a bracelet for my mother's birthday. I know that sounds pretty unremarkable, but what is remarkable is that I made it myself! I have had great fun designing and making the bracelet. (1) _____

My school has just started a new club called "The Workshop", where we are encouraged to make things ourselves. Students were invited to participate in the activity last week so I decided to give it a shot. (2) _____ However, I soon discovered that everybody just wanted to experiment and have fun making things. After designing the bracelet on the computer, I tried to make it with a three-D printer. (3) _____

But that wasn't a problem as everybody was just so helpful and supportive. In fact, the teacher in charge told us that if we didn't make mistakes, we probably weren't trying hard enough to be innovative and creative. That made us all laugh. No longer worrying about mistakes, I tried a couple more times and finally made it! Now I can't wait to see the look on my mother's face when I tell her I actually made the bracelet myself. (4) _____

- a My first attempts were a bit of a disaster, so I had to start again from scratch.
- b In fact, I'm still over the moon!
- c I was excited to learn being a maker was as easy as ABC.
- d It could take her breath away!
- e At first, I thought I would be like a fish out of water.

B3 What things have you made yourself or what would you like to make if you have the chance? Write a short paragraph about what you have made or would like to make. Use idioms where possible.

Integrated skills

Developing your creativity

A Colin is listening to a lecture about the relationship between experience and creativity. Listen and finish the exercises below.



A1 Listen to the lecture and choose the best answer. Circle the letters of the correct options.

- 1 What percentage of 10-year-old children scored at a genius level of creativity?
a 12%. b 30%. c 98%.
- 2 How was efficiency at the factory greatly improved in the early 20th century?
a By making workers move around machines.
b By building a block near the factory.
c By moving different parts of a product in front of workers.
- 3 What did James Watt do?
a He invented the steam engine.
b He improved the existing steam engine.
c He had a light bulb moment that came out of nowhere.
- 4 What is the attitude of the speaker to experience?
a Positive. b Negative. c Neutral.



A2 Listen to the lecture again and complete the notes below.

Experience and creativity

Experience discourages creativity

- The findings of a creativity test given to ⁽¹⁾ _____ and 280,000 adults suggest that as we grow older, we tend to lose ⁽²⁾ _____.
- Gaining experience and following set routines can sometimes lead to ⁽³⁾ _____.
- To get around mental blocks, we need to think out of the box and come at a problem ⁽⁴⁾ _____.

Experience encourages creativity

- In the words of Isaac Newton, standing on ⁽⁵⁾ _____ of Giants has enabled him to see further.
- A light bulb moment does not always come out of nowhere.
- A great innovation could be the result of ⁽⁶⁾ _____ and building on the work of previous generations.

B Colin has found an article giving tips on how to develop one's creativity. Read the article below and answer the following questions.



How to become more creative

In our ever-changing world, where new problems occur regularly and innovative solutions are always expected, creativity is becoming a skill increasingly in demand. Indeed, the value of creativity can never be overstated. In recognition of the important role that creativity and innovation play in all aspects of human development, the United Nations has been celebrating World Creativity and Innovation Day on 21 April annually since 2018.

Creativity is a skill which can be learnt. So, how can you become more creative? First of all, it has been well accepted that looking for connections inspires creativity. Things or ideas that seem unrelated are like dots in isolation, and when connected, they can form different pictures. This can be made easier by acquiring good subject knowledge, developing a broad understanding of human experience, and keeping a note of things you have observed as well as ideas you have come up with. The more dots you have, the more connections you are capable of making. Another means to improve creativity is to work with others. You don't have to be creative all on your own. As different people have different mindsets, skills and abilities, sharing ideas and combining their strengths would give rise to new perspectives, ideas and approaches, and ultimately lead to creative innovations. Last but not least, you must be ready to fight the fear of failure, which is often believed to be one of the biggest obstacles to creativity. In order to be creative, you need to be willing to take risks. Creativity is about experimenting and exploring. It is about trial and error. You need to accept that the process of innovation is a journey in which there will be both right and wrong turns. As long as you can reflect on the mistakes and make corrections accordingly, taking wrong turns is nothing to be feared.

Now that you know how to become more creative, what are you going to do about it? The simple answer is to start small. Creativity does not necessarily mean making a great discovery or invention; instead, it could mean making a minor improvement to an everyday situation or finding a simple solution to an everyday problem. Therefore, look for problems in your daily life that need to be solved. Try to find connections between different things and ideas, brainstorm with others, and don't be afraid to make mistakes. Then you will have your light bulb moment!

1 What tips are given on how to develop your creativity?

2 What can you do to be creative in your daily life?

C In pairs, have a discussion about creativity. Use the following questions to help you.

- What is creativity?
- Why is creativity important?
- What discourages creativity?
- How can we develop our creativity?

D Write an article about creativity. Use your ideas from part C and the information in parts A and B to help you. Check and improve your writing after you finish it.

	Title: _____
Introduction to creativity	_____

Reasons for lack of creativity	_____

Ways to develop creativity	_____

Conclusion	_____

Extended reading



Read the magazine article about how eyeglasses were invented.

How the world got eyeglasses



What are the greatest breakthroughs humans have made since the invention of the wheel? When a magazine posed this question to a panel of experts, including scientists, entrepreneurs, engineers and historians, eyeglasses came in behind only modern electronics, penicillin, electricity and the printing press.

- 5 If you disagree with this ranking, try this quick experiment: rub some soap on a piece of glass and hold it in front of your eyes. Or, if you wear glasses or contact lenses, remove them and try to keep reading. Oh, wait—you can't, because you can't see clearly enough. As anyone who needs corrective lenses will tell you, being able to see clearly is hugely important, whether you're trying to put together a set of building blocks or read your
- 10 favourite book.

- Considering that many people in the world suffer from poor vision, you may think that the great minds of history would have quickly come together to tackle the problem. However, this isn't at all what happened. Long before anything like spectacles existed, people had noticed that a piece of curved glass could magnify an image. About 2,000
- 15 years ago, people began making magnifying glasses this way. These were probably the first simple corrective lenses.

- No one knows who came up with the technology of shaping glass to bring images into better focus, or mounting lenses in frames that could go in front of your face. Most likely, many anonymous glassmakers tried to improve the technology by trial and error.
- 20 Although we don't know exactly whom to thank, it is pretty much agreed that corrective lenses first appeared in Italy in the 1280s.

- Once the technology was available, spectacles caught on quickly. Just like so many popular accessories today, spectacles soon became fashionable. People thought glasses made them look more intelligent or sophisticated. They were given away as gifts and
- 25 were popular among people who didn't even need them. In the 15th century, the printing press was invented, making books plentiful and relatively inexpensive. This drove the demand for reading glasses. There was, however, a practical problem to overcome. While the quality of the lenses was improving rapidly, getting them to stay comfortably

on the face was a great challenge. Indeed, reading while eating often meant spectacles in
30 the soup!

People made many attempts to overcome this problem, some stranger than others. Spectacles were held on the face by tying threads to the frames and looping them over the ears. Weights were sometimes tied to the ends of the threads to counterbalance the weight of the glasses to keep them from slipping off. Some people used threads or wires
35 to hook their glasses to their hats so that they dangled just in front of the eyes. Others attached them to straps tied around the forehead or threaded them into their hair. Still other designs simply perched uneasily on the bridge of the nose. Later models worked a bit better by having the frames slightly pinch the nose. These stayed on the face longer, but they must have been quite uncomfortable.

40 It's hard to imagine why it took so long to design spectacles with arm pieces that could clip behind the ears, but these weren't invented until the 18th century. Even then, the side stems were short and designed to clamp tightly against the temples (Ouch!). Like inventing the lenses themselves, coming up with designs that were easy to use was mostly a matter of trial and error. These days, technology has made spectacles
45 comfortable.

Unfortunately, now that we've tackled the technological problem of helping people see clearly, we face an ethical dilemma posed by the development of smart glasses, which are wearable computers in essence. Looking like a pair of ordinary glasses, smart glasses make it possible for the wearer to use the Web and make videos of everything
50 they see. However, opponents of smart glasses believe that a device that allows its wearer to record people without their knowledge, almost anywhere and anytime, is disturbing. Even though we have become used to the marvels of modern technology, we still need to pause and think about the potential negative impact of this new invention.

We'll have to wait and see if smart glasses would follow the same path as regular glasses,
55 or in other words, if their usefulness would outweigh potential problems. Meanwhile, those of us who need corrective lenses to read about the controversy should be grateful to that anonymous Italian glassmaker who first invented ordinary old spectacles.

A Write a summary of the magazine article. Use the following ideas to help you.

- The development of lenses
- The development of frames
- The ethical dilemma of smart glasses

B What do you think of the language feature of the magazine article? Use details from the article to support your opinion.

C What are mankind's most important inventions in your opinion? Make a list of the top five and explain briefly why they are important.

Project

Making a leaflet about an innovation

A As a class, discuss different innovations you are interested in. You can use the ideas below or think of other ideas. Then in groups, choose one innovation to research.

paper
compasses

electricity
steam engines

computers
batteries

wheels
the Internet

B As a group, research your chosen innovation. Use the questions below to help you.

- What was the problem?
- How was the problem solved?
- Why is this innovation important?
- What can we learn from it?

C As a group, put together your information to make your leaflet. Use the example below to help you. Then present your leaflet to the rest of the class.

Movable-type printing

Problem

Before the invention of movable-type printing, block printing was mainly used in China, which had obvious disadvantages:

- Time-consuming to make engraved blocks
- Inconvenient to store large numbers of engraved blocks
- Hard to correct engraving mistakes

Solution

In the Northern Song Dynasty, Bi Sheng invented movable-type printing, which was superior to block printing in many aspects:

- Small blocks could be put together freely, greatly reducing the plate-making time.
- Small blocks could be stored more easily, taking up less space.
- Small blocks could be disassembled and reused.

Significance

Movable-type printing has had major influences both at home and abroad:

- Improving the efficiency of printing
- Promoting mass literacy
- Accelerating the spread of knowledge and the progress of human civilization

Reflection

There is a lot to learn from the invention of movable-type printing:

- The wisdom of our ancestors has contributed to the welfare of mankind, which all Chinese can take pride in.
- Innovation is often driven by real-life needs.



Assessment

Complete the following learning journal to assess your performance. Then work in groups and exchange your learning journal with your partners.

Learning journal

In this unit I learnt _____

_____.

I was most interested in _____

_____.

I did well in _____

_____.

However, there is something I need to improve on: _____

_____.

In order to make more progress, I will _____

_____.

* Assess your learning of vocabulary and grammar by doing language practice on pages 62–63.

Further study



Creativity can be defined as something that has originality and functionality. It enables us to come up with new ideas and work out solutions to problems in everyday life. Creativity is highly prized in the modern world. Watch a documentary to find out more about what lies behind creativity.



What are humankind's greatest inventions? While there can be many different answers to this question, we all probably agree that without great inventions, our life would be entirely different from what it is now. Read a book on important inventions to learn how they have changed the world.



UNIT 3

Our culture, our treasure

In the long history, traditional Chinese culture, with its unique charm, has won praise from the world, and has been the pride of the Chinese people!

—Ji Xianlin



In this unit, you are going to:

- read a feature article about Chinese characters;
- write an article to promote Chinese shadow puppetry;
- read a magazine article about Confucius's philosophy;
- make a proposal for the school Chinese Culture Festival.

Welcome to the unit



China's rich culture is represented by a variety of unique symbols. Watch the video and finish the following exercises.



Chopsticks: fostering etiquette and taboos of their own, creating a bond between food and family, and passing on _____

The giant panda: biologically unique to China, one of the world's rarest wild species, and a symbol of _____

Beijing Opera: absorbing the essence of _____ and being highly appreciated for the artistic value of *Lianpu* make-up

- 1 Which of these symbols is your favourite? Why?
- 2 What other symbols of Chinese culture can you think of? Explain one of them briefly.

Reading



The Chinese writing system, based on Chinese characters, has played a crucial role in our culture and society. The feature article below is about Chinese characters. Before you read the article, think about the following questions:

- What do you know about the development of Chinese characters?
- What other writing systems are you familiar with?



Chinese characters: foundation of a remarkable writing system

In the year 1899, some mystical symbols carved on oracle bones attracted the attention of a scholar in Beijing. That was when people first began to realize the significance of these beautiful inscriptions. Since then, the mystery surrounding the origin of Chinese characters, the foundation of a remarkable writing system, has started to be revealed.

- 5 These oracle bone inscriptions, the earliest form of Chinese characters ever found to this day, are clear proof that the history of the Chinese writing system stretches back about 3,500 years. The inscriptions are highly complex, suggesting that earlier forms are yet to be discovered. All we have at the moment, however, is the legend of Cang Jie, who is said to have invented the first characters about 5,000 years ago. Legend has it that he
- 10 saw animal tracks in the ground, and began drawing pictures of objects in nature, such as the moon, the sun and the trees.

- Not surprisingly, the earliest Chinese characters were the fruits of our ancestors' careful observation and striking imagination. The Chinese character *shan*, for example, is simply a drawing of a mountain. However, our ancestors found not everything was
- 15 convenient to draw as a concrete image, so another way of forming Chinese characters was developed to represent abstract concepts. Some characters were created by adding symbols like a dot or a line to an existing pictographic character to indicate things. For instance, the characters *shang* and *xia* indicate things on top of and below a line. Gradually, these characters were far from enough to describe the colourful and diverse
- 20 world, so a new way of forming characters was developed by putting together different characters to indicate a new meaning. An example of this is the character *ming*, which combines *ri* for "sun" and *yue* for "moon" to express the meaning of being bright. These types of characters indicate meaning, but they do not provide clues about their pronunciation. Another group of characters were created by combining meaning and

25 pronunciation. In *hu*, the Chinese character for “lake”, for instance, the left part indicates the meaning and the right part the pronunciation. This method made it easier to create new characters, so most Chinese characters today belong to this group.

Since their creation, Chinese characters have been evolving consistently, increasing in practical value as a communicative tool. About 1,000 years after the oracle bone
30 inscriptions were used during the Shang Dynasty, Qinshihuang unified China and *Xiaozhuan* was made the official writing style of the Qin Dynasty. Then *Lishu* became a mainstream style of writing in the Han Dynasty. The evolution from *Xiaozhuan* to *Lishu* marked a revolutionary transformation from ancient characters into modern characters. Another milestone in the evolution of the style of Chinese characters was *Kaishu*,
35 which reached its peak in the Tang Dynasty. It became popular for its ease of use, and eventually progressed into what we now consider to be the standard script. Apart from changes in writing style, Chinese characters have been simplified so that they can be used more easily by the masses. Even in modern times, there have been attempts to simplify Chinese characters. One far-reaching change took place in the 1950s, when the
40 Chinese government simplified and standardized Chinese characters, promoting mass literacy.

Alongside their evolution, Chinese characters have sustained Chinese civilization over thousands of years. Thanks to Chinese characters, stories of our ancestors have been passed down in a form that remains accessible to us today, and the glorious history of
45 our country has been preserved. Moreover, Chinese characters have greatly influenced the writing systems in many other Asian countries such as Japan and Vietnam, forming the “Chinese character cultural sphere”. For instance, it was only after Chinese characters spread to Japan that the Japanese writing system started to be developed. With the spread of Chinese characters, educated people in these countries carefully studied
50 Chinese literary and philosophical texts, such as the works of the Tang Dynasty poets and *The Analects of Confucius*, and learnt the art of Chinese calligraphy. As we can see, Chinese characters hold the key to promoting the transmission of Chinese culture.

According to the famous writer Yu Guangzhong, ever since Chinese characters were born in ancient times, the memories and hopes of our ancestors have been well placed
55 in them. Indeed, Chinese characters provide an unbreakable link to our nation’s past and connect us to our ancestors. Every time we read or write Chinese characters, the whispers of our ancestors travel through time and space, and bring with them the stories of human civilization.